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Building a Personal Multi-Tenant Web Hosting Server with openSUSE Leap 16.0

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Yogyakarta | 4 October, 2026

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Building a Personal Multi-Tenant Web Hosting Server with openSUSE Leap 16.0

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BACKGROUND

Lightweight hosting without a traditional control panel

openSUSE 16

Explored as a lightweight alternative hosting platform.

Cockpit

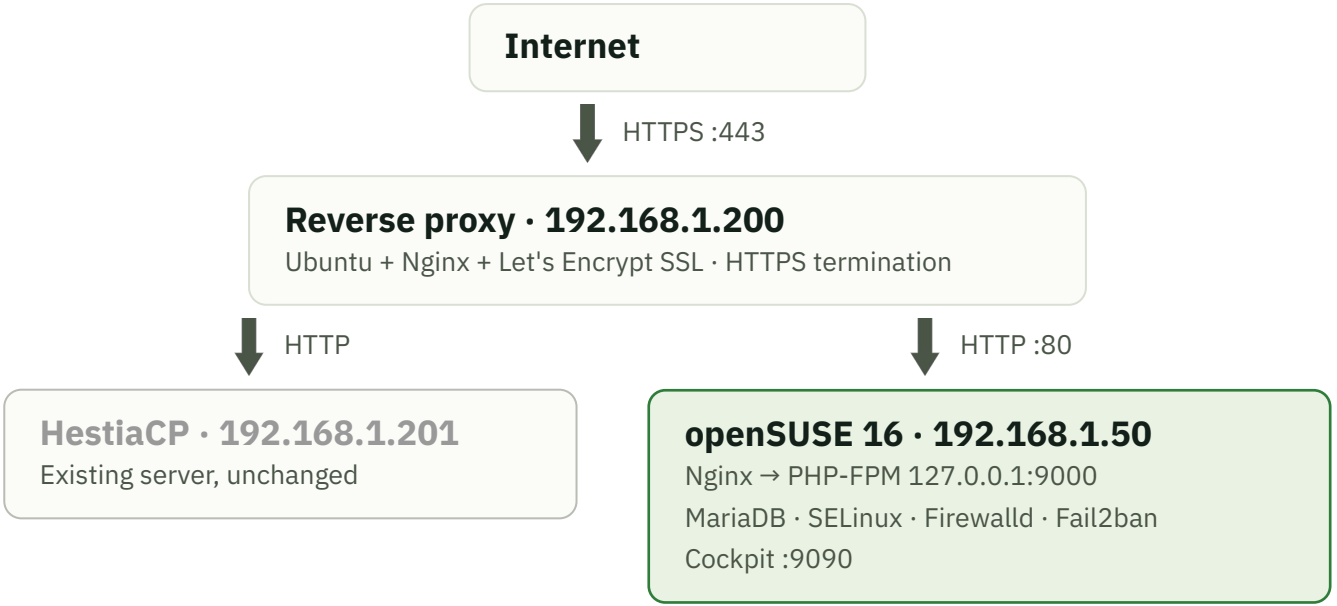
Serves as the web admin panel, replacing control panels such as HestiaCP.

Reverse proxy

The new server runs behind the existing Ubuntu proxy that already handles HTTPS.

ARCHITECTURE

HTTPS ends at the Ubuntu proxy



BEFORE YOU START

Six checks before the first command

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Know the server

openSUSE commands run as root on .50; proxy commands run with sudo on .200.

DNS first

Every domain needs an A record before Certbot. No www without its own record.

Plan the port

Each pool gets a unique port from 9001 that is also SELinux http_port_t.

Choose usernames

Lowercase letters and digits only; the name becomes the database prefix.

Protect secrets

Never paste passwords into chat or tickets; rotate any that leaked.

Snapshot first

Take a VM snapshot or back up /etc and databases before big changes.

Nine stages from install to verification

Stage	Component	Key configuration
1	Base install	openSUSE 16, static IP 192.168.1.50, SELinux Enforcing
2	Cockpit	Web admin panel on port 9090
3	Firewalld + Fail2ban	HTTP only from the proxy, Cockpit from LAN, sshd jail
4	MariaDB	WordPress database and user, localhost only
5	PHP 8.4 + PHP-FPM	php8 packages + php8-cli, listen 127.0.0.1:9000
6	Nginx	Vhost in vhosts.d + HTTPS FastCGI parameters
7	WordPress	restorecon, writable wp-content, wp-config checked with php -l
8	Reverse proxy	proxy_pass + X-Forwarded-Proto, Certbot certificate
9	Verification	curl, ausearch, DevTools free of Mixed Content

PROBLEMS FROM THE ARTICLE

Six initial problems and their root causes

Symptom	Cause	Fix
403 Forbidden	Vhost placed in conf.d	Move to /etc/nginx/vhosts.d
502 Bad Gateway	Assumed socket /run/php-fpm/www.sock	fastcgi_pass 127.0.0.1:9000
Access denied	SELinux label user_tmp_t from /tmp	restorecon -RFv /srv/www
Installer without CSS	HTTPS ends at the proxy	fastcgi_param HTTPS on, etc.
Mixed Content	WordPress builds http:// URLs	X-Forwarded-Proto + https WP_HOME
HTTP 500	Syntax error in wp-config.php	php -l wp-config.php

KEY CONFIGURATION

Two blocks that prevent the initial problems

Nginx vhost · openSUSE

```
location ~ \.php$ {
  fastcgi_pass 127.0.0.1:9000;
  fastcgi_param HTTPS on;
  fastcgi_param SERVER_PORT 443;
  fastcgi_param HTTP_X_FORWARDED_PROTO https;
}
```

The correct PHP-FPM port prevents 502; the HTTPS parameters prevent Mixed Content.

SELinux

```
restorecon -RFv /srv/www
setsebool -P httpd_can_network_connect on
semanage fcontext -a -t \
  httpd_sys_rw_content_t \
  "/srv/www/.../wp-content(/.*)?"
```

Correct labels prevent access denied without disabling SELinux.

WATCH-OUTS

Building the base server

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vhosts.d, not conf.d

openSUSE Nginx reads site configs from /etc/nginx/vhosts.d.

PHP-FPM is TCP

It listens on 127.0.0.1:9000 and ships only .default config files.

SELinux paths

Write rules for /var/www, run restorecon on /srv/www; never setenforce 0.

Pasting commands

A lost backslash splits a command. Paste heredocs whole, down to EOF.

Nothing above <?php

A blank first line breaks php -l and login cookies. Check every edit.

Test before reload

nginx -t and php-fpm -t first; on Ubuntu always with sudo.

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PART 2

From one site to multi-user hosting

Every user gets their own folder, PHP, databases and login

ACCOUNT MODEL

One Linux user equals one hosting account

User	Home = hosting folder	PHP-FPM pool	Databases
rhesa	/srv/www/rhesa/public_html	127.0.0.1:9001	wp_rhesa, rhesa_*
budi	/srv/www/budi/public_html	127.0.0.1:9002	budi_*

User homes live in /srv/www, so Cockpit Files opens straight into the user's own hosting folder. Create users with useradd -m -d /srv/www/<user> or set the default with useradd -D -b /srv/www.

REGISTRY

Keep one list of hosts, ports and accounts

Item	Value	Used by
Ubuntu proxy	192.168.1.200 :80 :443	SSL termination, routing-<domain> files
openSUSE backend	192.168.1.50 :80	Nginx, only reachable from the proxy
Cockpit	192.168.1.50 :9090	maintakraw.my.id, /files
PHP-FPM default pool	127.0.0.1:9000 (wwwrun:www)	phpMyAdmin
PHP-FPM pool rhesa	127.0.0.1:9001	rhesaprawedha.my.id
PHP-FPM pool budi	127.0.0.1:9002	budi.maintakraw.my.id
Next user	*127.0.0.1:9003*	*Check with ss and the pool files first*

ISOLATION

Three layers that separate users

Folder mode 2750

Owner is the user, group nginx, setgid. Other users cannot open the folder, not even through Cockpit.

PHP-FPM pool per user

PHP runs as that user and is locked by open_basedir to public_html and tmp.

Explicit SELinux labels

Folder, wp-content, tmp and logs labelled via /var/www rules; each pool port registered as http_port_t.

COCKPIT FILES

Cockpit opens straight into the user's folder

- Direct link per user: files#/?path=%2Fsrv%2Fwww%2Fbudi
- Bookmarks for public_html and logs next to Home
- Non-wheel users get Limited access; admin access asks for the root password
- Users change passwords with passwd in Cockpit Terminal

Install from the official release

```
V=34
curl -LO .../cockpit-files-$V.tar.xz
tar xJf cockpit-files-$V.tar.xz
cp -r cockpit-files/dist/* \
  /usr/share/cockpit/files/
restorecon -RFv /usr/share/cockpit
```

Not in the official Leap 16 repos. Needs Cockpit 318 or newer; the server runs cockpit-bridge 354.

WATCH-OUTS

Users, Cockpit and PHP pools

Create users by script

Cockpit Accounts gives no database, no pool and a /home folder.

Check the home

getent passwd <user> must show /srv/www/<user>.

Send the Files link

Use files#/?path=...; a cached URL can open / instead.

Root password prompt

Admin access uses sudo targetpw. Expected; never share root.

Password changes

Cockpit: passwd in Terminal. phpMyAdmin: Change password.

One pool stops all

Unregistered port = PHP-FPM down for every site and phpMyAdmin.

One phpMyAdmin, different content for every user

Login
User signs in with their own MariaDB account

→

Cookie auth
phpMyAdmin stores no users at all

→

GRANT
MariaDB decides the access rights

→

Result
Only that user's databases

rhesa sees
wp_rhesa and every rhesa_* database

budi sees
Only budi_* databases

root
Refused on the web because AllowRoot = false

PHPMYADMIN

Served at maintakraw.my.id/phpmyadmin

config.inc.php

```
$cfg['Servers'][$i]['auth_type'] = 'cookie';
$cfg['Servers'][$i]['AllowRoot'] = false;
$cfg['TempDir'] = '/var/lib/phpmyadmin/tmp';
$cfg['PmaAbsoluteUri'] =
  'https://maintakraw.my.id/phpmyadmin/';
```

PmaAbsoluteUri is required because phpMyAdmin sits in a subfolder behind the proxy.

Ubuntu proxy · maintakraw.my.id server

```
location /phpmyadmin/ {
  proxy_pass http://192.168.1.50/;
  proxy_set_header Host $host;
  proxy_set_header X-Forwarded-Proto $scheme;
}
```

The trailing slash on proxy_pass strips the /phpmyadmin prefix.

PART 3

Automation and proxy

. . .

Everyday commands for database users

Task	Command
Add a user	CREATE USER 'budi'@'localhost' IDENTIFIED BY '...';
Grant prefixed databases	GRANT ALL PRIVILEGES ON `budi\_%`.* TO 'budi'@'localhost';
Check rights	SHOW GRANTS FOR 'budi'@'localhost';
Reset password	ALTER USER 'budi'@'localhost' IDENTIFIED BY '...';
Revoke access	REVOKE ALL PRIVILEGES ON db_name.* FROM 'budi'@'localhost';
Delete a user	DROP USER 'budi'@'localhost'; (databases are kept)

Run FLUSH PRIVILEGES after GRANT or REVOKE. Users can change their own password from the Change password menu.

WATCH-OUTS

phpMyAdmin and MariaDB

SHOW GRANTS FOR
Without FOR it shows root's rights, not the user you created.

IF NOT EXISTS
Keeps an old password. Reset it with ALTER USER, then test.

Three accounts
Linux rhesa, MariaDB rhesa, wp_rhesa. Only wp_rhesa affects the site.

Inside server { }
Proxy location blocks go inside the listen 443 block.

PmaAbsoluteUri
Required in a subfolder, or login loops back to the form.

Prefix and underscore
Other databases need their own GRANT; _ is a wildcard.

AUTOMATION

add-hosting-user.sh creates a complete account

```
sudo ./add-hosting-user.sh budi budi.maintakraw.my.id 9002
```

0
Validate
Name, domain, port free in ss and every pool

1
User
Folders, bookmarks, SELinux labels and port

2
PHP-FPM
Own pool, open_basedir

3
Nginx
Vhost to public_html

4
MariaDB
GRANT on budi_*

5
Verify
php-fpm -t, service up, DB login works

Any failure triggers an automatic rollback in reverse order, and PHP-FPM and Nginx are restarted if they went down.

AUTOMATION

remove-hosting-user.sh undoes an account safely

```
sudo ./remove-hosting-user.sh budi --dry-run
```

Refuses risky targets

root, nos, system accounts, wheel members, homes outside /srv/www.

Backs up first

Files, configs and database dumps to /root/hosting-backups; stops if the backup fails.

Removes everything

Sessions, vhosts, pools, SELinux rules, budi_* databases, MariaDB and Linux user.

Databases outside the prefix, such as wp_rhesa, are kept; only the user's access is removed.

REVERSE PROXY

One template for every routing-<domain> file

routing-template.conf

```
server {
  listen 443 ssl;
  server_name {{DOMAIN}};
  ssl_certificate .../{{DOMAIN}}/fullchain.pem;
  location / {
    proxy_pass http://{{BACKEND}};
    proxy_set_header X-Forwarded-Proto https;
  }
}
server {
  listen 80;
  return 301 https://$host$request_uri;
}
```

HTTPS forwards to the backend; HTTP only redirects.

```
sudo ./add-proxy-route.sh budi.maintakraw.my.id
```

1. Checks DNS, backend answer and duplicate domains
2. Enables a temporary port 80 block
3. certbot certonly: Certbot never edits the file
4. Writes the final blocks from the template
5. nginx -t, reload, and verifies there is no redirect loop

Any failure restores the previous config.

WATCH-OUTS

Reverse proxy and certificates

Certbot once
Use certonly through the script; repeated --nginx runs corrupt files.

Right block, right job
443 holds proxy_pass; 80 only returns 301 to HTTPS.

Find the real source
The proxy rewrites Server; curl the backend with a Host header.

Cached redirects
Browsers keep 301s. Test in a private window after a fix.

Watch renewals
certbot renew --dry-run; check the domain had already expired.

Clean removals
Delete routing-<domain>, reload Nginx, certbot delete.

RUNBOOK

Adding a new site, step by step

#	Where	Command	Expected
1	DNS	dig +short <domain>	The public IP
2	openSUSE	sudo ./add-hosting-user.sh <user> <domain> <port>	OK: phpMyAdmin access ready
3	openSUSE	curl -s -H "Host: <domain>" http://127.0.0.1/	200 once index file exists
4	Ubuntu	sudo ./add-proxy-route.sh <domain>	Done: https://<domain>
5	Browser	https://<domain> in a private window	Page loads, no Mixed Content
6	User	Files link + phpMyAdmin from credentials file	Own folder, own databases

TROUBLESHOOTING

Work from the outside in

Browser
DevTools, private window

→

Proxy
curl --max-redirs 0, sudo nginx -t

→

Backend
curl -H Host to .50

→

PHP-FPM
is-active, ss -ltn, journalctl

→

SELinux / DB
ausearch, SHOW GRANTS FOR

Stop at the first layer that misbehaves. Each layer has one command that proves it is healthy before you move inward.

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PART 4

Field notes and wrap-up

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Cockpit and user accounts

Problem	Cause	Fix
cockpit-files not found	Not in the official Leap 16 repos	Copy dist/ from GitHub release v34
File browser not showing	Copied to /usr/local/share	Use /usr/share/cockpit/files
User home in /home	Created via Accounts without -d	usermod -m -d /srv/www/<user>
Files opens / instead of home	URL carries path=/ from earlier session	Direct path link + bookmarks
Password change asks for root	Admin access uses sudo targetpw	passwd in Cockpit Terminal

phpMyAdmin and database

Problem	Cause	Fix
php: command not found	PHP CLI not installed	php-fpm -m; install php8-cli
php -l: strict_types error	Blank line above <?php	Remove lines above <?php
location not allowed here	Block outside server { }	Move inside the listen 443 block
nginx -t Permission denied	Run without sudo	sudo nginx -t && reload
#1045 Access denied	Old password kept by IF NOT EXISTS	ALTER USER, test mariadb -u

SELinux and the scripts

Problem	Cause	Fix
semanage equivalency error	Policy treats /srv/www as /var/www	Write rules with /var/www/<user>
Half-created account	Old script had no rollback	Automatic rollback in reverse order
php-fpm cannot bind 9002	Port not http_port_t	semanage port -a -t http_port_t
Every site down at once	One failing pool stops all pools	php-fpm -t, check service, restart
Site not found after move	Old vhost root and pool	Update root and fastcgi_pass

FIELD NOTES 4/4

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Reverse proxy and certificates

Problem	Cause	Fix
ERR_TOO_MANY_REDIRECTS	443 block only redirected to itself	Rewrite from template; certbot certonly
certbot fails on www	www.budi... has no DNS record	Drop www, or add the record first
Redirect looked like Ubuntu	Proxy replaces the Server header	Test backend with Host header
Browser still loops after fix	301 cached by the browser	Private window or clear cookies
website.XX.id cert expired	Renewal not running for it	certbot renew --dry-run or delete

LESSONS LEARNED

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What to remember for the next server

Check distro conventions
vhosts.d, PHP-FPM port, php8 names, /srv/www = /var/www.

Label folders and ports
SELinux matters as much as file permissions.

One pool can stop all
Test with php-fpm -t and check after every reload.

Forward HTTPS info
WordPress and phpMyAdmin both need it behind a proxy.

Official upstream releases
The answer when Leap 16 lacks a package.

Automate with rollback
Validate, verify, and never leave half-made accounts.

PLANS

Toward oS-Hosting on top of Cockpit

Feature	Foundation in place	Status
Multi-user hosting	Linux users + PHP-FPM pool per user	Running
Database Management	phpMyAdmin + prefixed GRANTs	Running
Account automation	add- and remove-hosting-user.sh	Running
Domain Management	Backend vhost + add-proxy-route.sh	Running
SSL Management	certbot certonly in add-proxy-route.sh	Running
Backup and Restore	Backup archive in the remove script	Partial
WordPress Installer	Stage 7 steps	Planned
Role Based Access	Linux groups recognized by Cockpit	Planned

Thank You



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